

SQL CHEATSHEET

CATEGORIES

Category	Full Name	Description
DDL	Data Definition Language	Commands that define database structure
DQL	Data Query Language	Commands for retrieving data
DML	Data Manipulation Language	Commands that manipulate data
DCL	Data Control Language	Commands for access control
TCL	Transaction Control Language	Commands for transaction management

DATABASE OBJECTS

TABLE	Stores data in rows and columns
VIEW	Virtual table based on SQL statement
SYNONYM	Alternative name for an object
SEQUENCE	Generates numeric values
INDEX	Improves query performance
TRIGGER	Automated action on table events

SQL KEYWORDS

TYPE	Keywords
Important Keywords	WHERE, DISTINCT, LIMIT, ORDER BY, DESC, ASC, AS, FROM, SET, VALUES, CASE, DEFAULT
Aggregation Keywords	GROUP BY, HAVING

SQL COMMANDS BY CATEGORY

DDL CREATE, DROP, ALTER, TRUNCATE, RENAME, COMMENT

BASIC DDL EXAMPLES

Create a Table

```
CREATE TABLE Students(rollno int PRIMARY KEY, name varchar(30) NOT NULL, age int)
```

Add a Column

```
ALTER TABLE Students ADD email varchar(100)
```

Modify Column

```
ALTER TABLE Students MODIFY email varchar(200)
```

Remove Column

```
ALTER TABLE Students DROP COLUMN email
```

Drop a Table

```
DROP TABLE Students
```

DQL SELECT

BASIC DQL EXAMPLES

Fetch all records

```
SELECT * FROM Students
```

Filter data

```
SELECT * FROM Students WHERE rollno=234
```

Sort records

```
SELECT name, grade FROM Students ORDER BY gender
```

Join tables

```
SELECT * FROM Students INNER JOIN Section ON Students.section_id = Section.id
```

DML INSERT, UPDATE, DELETE, LOCK, CALL, EXPLAIN PLAN

BASIC DQL EXAMPLES

Insert data

```
INSERT INTO Students(rollno, name, age) VALUES (234, 'Donald', 21)
```

Update data

```
UPDATE Students SET age=22 WHERE rollno=234
```

Delete data

```
DELETE FROM Students WHERE rollno=234
```

Aggregate

```
SELECT avg(age) AS AverageAge FROM Students GROUP BY section
```

DCL

GRANT, REVOKE

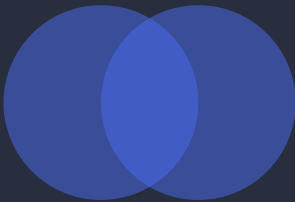
TCL

COMMIT, ROLLBACK, SAVEPOINT, SET TRANSACTION

SET OPERATIONS

UNION

Combines results, removes duplicates

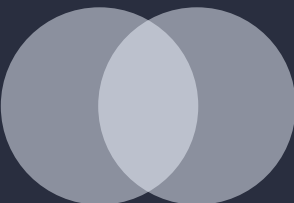


UNION ALL

Combines results, keeps duplicates

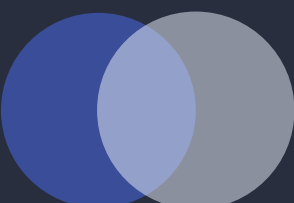
INTERSECT

Returns common rows



EXCEPT/MINUS

Returns rows in first query but not in second



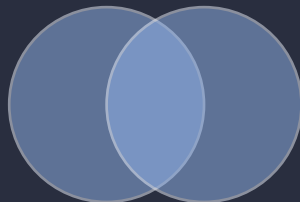
CONSTRAINTS

PRIMARY KEY	Virtual table based on SQL statement
FOREIGN KEY	Links tables together
UNIQUE	Ensures all values are different
CHECK	Ensures values meet conditions
DEFAULT	Provides default value
NOT NULL	Ensures column cannot be empty

JOINS

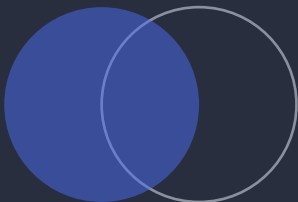
INNER JOIN

Returns matching rows from both tables



LEFT [OUTER] JOIN

Returns all rows from left table and matching from right



FULL [OUTER] JOIN

Returns all rows when there's a match in either table



RIGHT [OUTER] JOIN

Returns all rows from right table and matching from left



SQL OPERATORS

ARITHMETIC

+, -, *, /, %

BITWISE

&, |, ^, ~

LOGICAL

AND, OR, NOT, ANY, SOME, ALL, BETWEEN, IN, EXISTS, LIKE, NULL, UNIQUE

COMPARISON

=, >, <, ≥, ≤, >, <, ≠

COMPOUND

+=, -=, *=, /=, %=, &=, ^=, |=

AGGREGATION FUNCTIONS

AVG

Returns average value

COUNT

Returns number of rows

MAX

Returns maximum value

MIN

Returns minimum value

SUM

Returns total sum